Facial Expression Classification

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I classified 48x48 pixel grayscale faces into 3 emotions--angry, happy, and neutral--by fine-tuning a ResNet-50, training only its final residual block and a new three-class classifier. I used a stratified train-validation split and data augmentation to limit overfitting. The model achieved ~84% accuracy on the validation set and ~83% accuracy on the test set.

METHOD

Preliminary data analysis (see Figure 1) revealed a class imbalance: happy faces (labeled 1) substantially outnumbered neutral (2) and angry (0) faces. I split the data with stratification so that the training and validation sets preserved these class proportions, making validation accuracy a representative estimate of test performance. Because Kaggle scores accuracy, which can be inflated by predicting the majority class, I also monitored the precision, recall and f1 scores to confirm that the model was not defaulting to the happy class.

An initial model (see Figure 2) used a class-balanced sampler to counter class imbalance; the final model trained on the natural class distribution, since the evaluation metric is accuracy.

After running the model and observing the evaluation metrics (see Figure 2), I concluded that overfitting was harming my model performance. To mitigate this issue, I added a nn.Dropout component to the fully connected layer of the model, added label smoothing to the criterion, and added per-epoch evaluation to catch overfitting early.

I originally used a learning rate of 1e-4 for all layers. However, my best loss was at Epoch 1, and it worsened from there (see Figure 2). This suggests that my learning rate was too large. I decreased the learning rate overall and customized it per-layer.

The single most advantageous adjustment I made was resizing the images to be 96x96 pixels (3x3 grid). I initially set the size to be 48x48 (2x2 grid), which evidently was a lower resolution than the images used to train the early layers of the ML model. Resizing the images increased the accuracy of the test set predictions by ~11 points and generally improved all metrics.

```
Validation set metrics:

Epoch 1: Loss 0.7958, Accuracy 0.640, Precision 0.624, Recall 0.628, f1 0.622
--> New best model saved! (Best F1: 0.622)
Epoch 2: Loss 0.8125, Accuracy 0.659, Precision 0.644, Recall 0.647, f1 0.644
--> New best model saved! (Best F1: 0.644)
Epoch 3: Loss 0.8913, Accuracy 0.664, Precision 0.649, Recall 0.644, f1 0.645
--> New best model saved! (Best F1: 0.645)
Epoch 4: Loss 1.0599, Accuracy 0.664, Precision 0.641, Recall 0.632, f1 0.635
Epoch 5: Loss 1.1083, Accuracy 0.655, Precision 0.639, Recall 0.630, f1 0.632
Epoch 6: Loss 1.2653, Accuracy 0.658, Precision 0.639, Recall 0.640, f1 0.639
Epoch 7: Loss 1.3700, Accuracy 0.655, Precision 0.639, Recall 0.643, f1 0.640
Epoch 8: Loss 1.4124, Accuracy 0.675, Precision 0.656, Recall 0.654, f1 0.654
--> New best model saved! (Best F1: 0.654)
Epoch 9: Loss 1.5164, Accuracy 0.659, Precision 0.640, Recall 0.644, f1 0.641
Epoch 10: Loss 1.6425, Accuracy 0.658, Precision 0.643, Recall 0.631, f1 0.634
```

Figure 2

```
train_target_df.value_counts()

0
1    7215
2    4965
0    3995
Name: count, dtype: int64
```

Figure 1

Acknowledgements

I finetuned Ziad Nasser's model, as his model was successful in the Kaggle competition Challenges in Representation Learning: Facial Expression Recognition Challenge. This challenge served as the inspiration for the Kaggle competition Facial Expression Classification—the crux of this project. Notably, Ziad Nasser's model uses ResNet-50 architecture, modified for single-channel (grayscale) images instead of standard 3-channel RGB images. ResNet-50 architecture is the industry standard for image classification tasks.

INTRODUCTION

Automatic facial recognition supports applications such as human-computer interaction and accessibility tools. Creating a model suited to this task is difficult because facial expressions can be subtle, with similar expressions conveying vastly different emotions. Models trained on limited data tend to disregard less-common emotions or otherwise misclassify certain expressions—particularly in the case of ethnic minorities. This project asks how well a pretrained network can be adapted to recognize three emotions (angry, happy, neutral), and which choices in adapting it matter most.

RESULTS

My best model achieved an accuracy score of ~84%, a precision score of ~83%, a recall score of ~82%, and an f1 score of ~82% (see Figure 3). This is a ~8 point improvement from the baseline accuracy of ~76%.

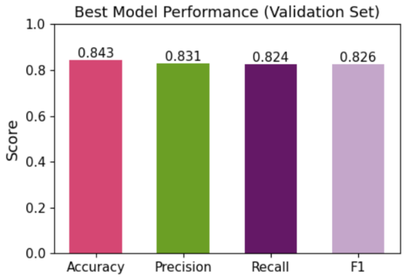


Figure 3

CONCLUSION

The changes I made that yielded the greatest improvements in accuracy were accounting for overfitting, decreasing the learning rate, and resizing the images to be larger.

Future work could entail increasing the number of training epochs and expanding the classification task to recognize a wider range of emotions.

This model could be significantly improved if trained on more data. Ideal training data would showcase an equal sample of a wide variety of emotions. It is also pertinent that the data would showcase great variation in age, gender, and ethnicity; an optimal dataset would showcase an equal number of participants across these demographics to mitigate performance gaps associated with minority groups.

REFERENCES

- Data Augmentation in Machine Learning: Improving Model Performance
- Simple Image Classification with ResNet-50

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